Statistics
Lecture 08
Two-sample hypothesis testing

General recipe for hypothesis testing (review)

Case: Non-paired test for difference in means for two normal distributions with known variances

- **Model**
- **Hypothesis: 2-sided/1-sided**
- **Estimators and test statistic for difference in means and its distribution**
- **Test statistic Z_0**
- **Hypothesis test for difference in means using α approach or p-value approach**
- **Confidence interval for difference in means**
- **(Type 2 error probabilities and power)**

Case: Non-paired test for difference in means for two normal distributions with unknown but equal variances (MATLAB: ttest2)

- **Test statistic Z_0 using equal variances**
- **Estimators for common variance and pooling of estimators**
- **Test statistic T_0**
- **Hypothesis test for difference in means using α approach or p-value approach**
- **Confidence interval for difference in means**

Case: Non-paired (large sample, approximate) test for difference in means for two normal distributions with unknown and unequal variances (MATLAB: ttest2)

- **Test statistic Z_0^***

Case: Paired test for difference in means for two normal distributions (MATLAB: ttest)

- **Model and difference model**
- **Test statistic T_0**

Case: Test for equal variances for two normal distributions (MATLAB: vartest2)

- **Hypothesis**
- **Estimators for variances**
- **Test statistic, F_0**
- **The Fisher F-distribution**
- **Hypothesis testing and confidence interval for ratio of variances**